

ASSIGNMENT-02

Harshini R N

192421175

CSA5117

Cryptography and Network Security

94
100

Q: Blowfish - S-Box 1 in F Function

Given i/p:

01101001

01101001

01101001

Each 01101001 is:

$$01101001_2 = 105_{10}$$

In blowfish the i/p to the F-function is normally 32 bits divided into four 8-bit parts.

$$x = a || b || c || d$$

The first 8 bits are used as the index for S-box

So,

$$a = 01101001 = 105$$

$$S_1[105]$$

The F-function then uses all four S-boxes

$$F(x) = [(S_1[a] + S_2[b]) \oplus S_3[c]] + S_4[d]$$

2) Diffie Hellman Key exchange

Given:

$$g = 8, p = 23$$

Alice's private Key:

$$a = 6$$

Bob's private key:

$$b = 15$$

Alice public key

$$A = g^a \text{ mod } p$$

$$A = 5^6 \text{ mod } 23 = 8$$

Bob's public key

$$B = g^b \text{ mod } p$$

$$B = 5^{15} \text{ mod } 23 = 19$$

Shared Key

Alice calculates:

$$K = B^a \text{ mod } p$$

$$K = 19^6 \text{ mod } 23 = 2$$

Bob calculates:

$$K = A^b \text{ mod } p$$

$$K = 8^{15} \text{ mod } 23 = 2$$

$$\therefore \text{shared Key} = 2$$

3. Blowfish vs RC5 VS DES

BLOWFISH:

Key size: 32-448 bits (flexible)

Structure: 64-bit blocks, 16-round Feistel, key dependent S-boxes.

Performance: Fast after setup, slower if keys change often.

RC5:

Key Size: 0-2040 bits highly customizable.

Structure: Variable block size (32/64/128 bits), rounds (0-255), uses rotations, XOR, addition.

Performance: Speed / security depends on chosen parameters, good for flexible systems.

DES:

Key Size:- Fixed 56 bits (weak today)

Structure: 64-bit blocks, 16 round Feistel, fixed S-boxes.

Performance: Slower and insecure by modern standards.

4. Block Cipher vs stream cipher Block Cipher

Block Cipher

Encrypts data in fixed-size blocks

Example: AES, Blowfish, DES

Used for files, databases etc...

Example:

A 128 bit block of a file $\rightarrow$ encrypted
 $\rightarrow$ Cipher text block:

Stream Cipher

Encrypts data continuously, usually bit-by-bit
or byte-by-byte.

Example: ChaCha20

Useful for real time communication

Basic formula

$$C = P \oplus K$$

where,

P = plaintext

K = Key stream

C = Cipher text.

Simple Difference

Blocks = fixed blocks

Stream = Continuous data

Importance of Key Management in public-key cryptography

public key Systems use:

- Public Key: Can be shared
- Private Key: must be kept secret

Why Key management is important?

1. Protects private keys from the attackers
2. Ensures the correct public key that belong to the correct person / website
3. Prevents Impersonation.
4. Allows keys to be renewed or revoked when compromised.

→ That is why key management in public key cryptography is important.

Digital Certificate

A digital certificate connects

Identity + Public Key

A certificate Authority (CA) verifies the identity and digitally signs the certificate.

Example

When visiting a banking website, the browser checks its certificates to verify that the public key really belongs to that website.

→ If a private key is stolen, the certificate can be revoked.

Main Point

Good key Management =
Confidentiality + authentication
+ security.

Q